문제 3. 일반 상대성 이론의 기초 (Fundaments of General Relativity) — 해설

원문: Romanian Master of Physics 2012, PROBLEM No. 3 해설. 배점은 원문대로 pt로 표기한다.

a. **0.5 pt**

로런츠 변환(Lorentz transformation)을 두 번 미분하면 다음을 얻는다.

$$a_x = a'_x \left(\frac{\sqrt{1-\frac{u^2}{c^2}}}{1+\frac{u}{c^2}v'_x} \right)^3$$

우리의 경우 $u = v_x$이고 $v'_x = 0$이다.

$$\frac{\mathrm{d}v_x}{\mathrm{d}t} = a' \left(1-\frac{v_x^2}{c^2}\right)^{\frac{3}{2}}$$

$$F_x = \frac{ma_x}{1-\frac{v_x^2}{c^2}} = m_0 a' = \text{constant}$$

b. **0.5 pt**

$$v_x = c\sin\alpha \Rightarrow \frac{\mathrm{d}\left(c\sin\alpha\right)}{\left(1-\sin^2\alpha\right)^{\frac{3}{2}}} = a'\mathrm{d}t \Rightarrow c\tan\alpha = a't + C$$

$t = 0$에서 $v_x = 0$이므로 $\alpha = 0$이고 $C = 0$이다.

$$\frac{\frac{v_x}{c}}{\sqrt{1-\frac{v_x^2}{c^2}}} = \frac{a't}{c} \Rightarrow v = c\,\frac{\frac{a't}{c}}{\sqrt{1+\left(\frac{a't}{c}\right)^2}}$$

c. **0.5 pt**

$$\mathrm{d}t' = \mathrm{d}t\sqrt{1-\frac{v^2}{c^2}} = \frac{\mathrm{d}t}{\sqrt{1+\left(\frac{a't}{c}\right)^2}}; \quad \frac{a't}{c} = \sinh\tau \Rightarrow \mathrm{d}t' = \frac{c}{a'}\,\mathrm{d}\tau \Rightarrow t' = \frac{c}{a'}\tau + C$$

다시 $C = 0$이므로

$$t' = \frac{c}{a'}\,\mathrm{arcsinh}\left(\frac{a't}{c}\right)$$

d. 1 pt

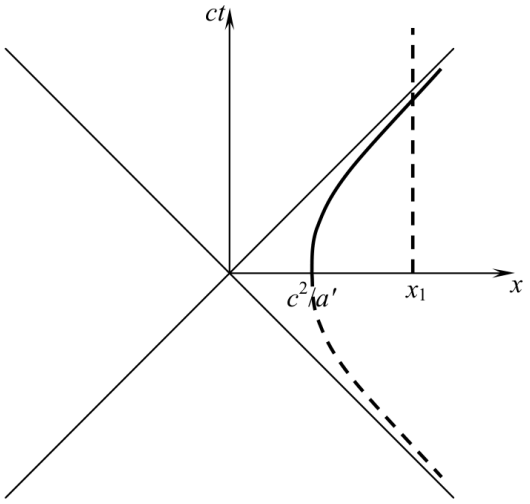
$$\left.\begin{aligned} -c^2\left(\mathrm{d}t'\right)^2 &= -c^2\left(\mathrm{d}t\right)^2 + \left(\mathrm{d}x\right)^2 \\ \frac{a't}{c} &= \sinh\tau \Rightarrow \mathrm{d}t = \frac{c}{a'}\cosh\tau\,\mathrm{d}\tau \end{aligned}\right\} \Rightarrow \left(\mathrm{d}x\right)^2 = \frac{c^4}{a'^2}\left(\cosh^2\tau - 1\right)\left(\mathrm{d}\tau\right)^2 \Rightarrow$$

$$dx = \frac{c^2}{a'}\sinh\tau\,\mathrm{d}\tau \Rightarrow x = \frac{c^2}{a'}\cosh\tau + C$$

$t = t' = 0$에서 $x_0 = c^2/a'$이므로 다시 $C = 0$이다.

e. 1 pt

$$ct = \frac{c^2}{a'}\sinh\tau \Rightarrow x^2 - \left(ct\right)^2 = \left(\frac{c^2}{a'}\right)^2 \Rightarrow \frac{x^2}{\left(\frac{c^2}{a'}\right)^2} - \frac{\left(ct\right)^2}{\left(\frac{c^2}{a'}\right)^2} = 1$$



f. 0.5 pt

$$\rho_0 = \frac{c^2}{a'} \Rightarrow \begin{cases} x = \rho_0\cosh\tau \\ ct = \rho_0\sinh\tau \end{cases}$$

g. 0.5 pt

$$\begin{cases} x = \rho\cosh\tau \\ ct = \rho\sinh\tau \end{cases}; \qquad \begin{cases} \rho = \sqrt{x^2 - \left(ct\right)^2} \\ \tau = \operatorname{arctanh}\left(\frac{ct}{x}\right) \end{cases}$$

이 방정식들은 $x > 0$이고 $\rho > 0$일 것을 요구하므로, 이 새로운 매개변수들로는 $x > |ct|$로 특징지어지는 시공간의 사분면만을 덮을 수 있다.

h. 1 pt

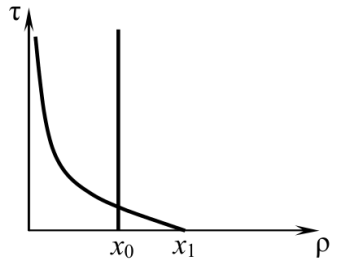
$$\left.\begin{aligned} \mathrm{d}x &= \mathrm{d}\rho\cosh\tau + \rho\sinh\tau\,\mathrm{d}\tau \\ \mathrm{d}(ct) &= c\,\mathrm{d}t = \mathrm{d}\rho\sinh\tau + \rho\cosh\tau\,\mathrm{d}\tau \end{aligned}\right\} \Rightarrow$$

$$ds^2 = -c^2\left(\mathrm{d}t\right)^2 + \left(\mathrm{d}x\right)^2 = \left(\mathrm{d}\rho\right)^2 - \rho^2\left(\mathrm{d}\tau\right)^2 = -c^2\frac{\rho^2}{c^2}\left(\mathrm{d}\tau\right)^2 + \left(\mathrm{d}\rho\right)^2; \quad f = \frac{\rho^2}{c^2}; \quad g = 1$$

i. 0.5 pt

$$\rho = \frac{x_1}{\cosh\tau} \Leftrightarrow \tau = \operatorname{arccosh}\left(\frac{x_1}{\rho}\right)$$

$$\Delta\rho = \frac{c^2}{a'}$$



j. 0.5 pt

관측자는 신호기(beacon)가 린들러 계량(Rindler metric)으로 기술되는 시공간 사분면을 벗어나기 전에 방출된 신호만을 받는다.

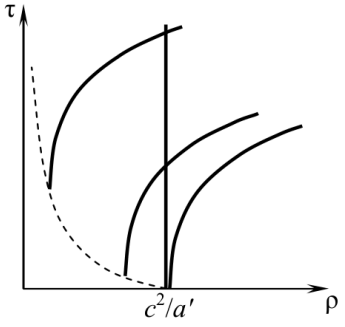
$$ct \le x = x_0 \Rightarrow t_{\lim} = \frac{x_0}{c} \Rightarrow N = \left[\frac{t_{\lim}}{T_0}\right] + 1 = \left[\frac{x_0}{cT_0}\right] + 1$$

빛의 경우,

$$ds^2 = 0 \Rightarrow (\mathrm{d}\rho)^2 = \rho^2 (\mathrm{d}\tau)^2 \Rightarrow \frac{\mathrm{d}\rho}{\rho} = \mathrm{d}\tau$$

$\tau = 0$에서 $\rho_0 = x_0$이므로

$$\ln\frac{\rho}{\rho_0} = \tau \Rightarrow \rho = x_0\,\mathrm{e}^{\tau}$$



k. 1.5 pt

펄스가 방출되는 시공간 좌표를 ρ_e, τ_e라 하자.

$$\rho_\mathrm{e} = \sqrt{x_0^2 - c^2t^2}\,; \quad \tau_\mathrm{e} = \operatorname{arctanh}\left(\frac{ct}{x_0}\right)$$

$$\tanh\tau_\mathrm{e} = \frac{\mathrm{e}^{\tau_\mathrm{e}} - \mathrm{e}^{-\tau_\mathrm{e}}}{\mathrm{e}^{\tau_\mathrm{e}} + \mathrm{e}^{-\tau_\mathrm{e}}} = \frac{ct}{x_0} \Rightarrow \mathrm{e}^{2\tau_\mathrm{e}} - 1 = \frac{ct}{x_0}\left(\mathrm{e}^{2\tau_\mathrm{e}} + 1\right) \Rightarrow \mathrm{e}^{2\tau_\mathrm{e}} = \frac{1 + \dfrac{ct}{x_0}}{1 - \dfrac{ct}{x_0}} \Rightarrow \mathrm{e}^{\tau_\mathrm{e}} = \sqrt{\frac{x_0 + ct}{x_0 - ct}}$$

$$\rho = \frac{\rho_\mathrm{e}}{\mathrm{e}^{\tau_\mathrm{e}}}\,\mathrm{e}^{\tau} = (x_0 - ct)\,\mathrm{e}^{\tau} = \rho_0 \Rightarrow \mathrm{e}^{\tau} = \frac{x_0}{x_0 - ct} \Rightarrow \tau = \ln\left(\frac{x_0}{x_0 - ct}\right)$$

관측자가 마지막 신호를 받는 순간을 t^*라 하자.

$$v(t^*) = c\,\frac{\dfrac{a't^*}{c}}{\sqrt{1 + \left(\dfrac{a't^*}{c}\right)^2}}$$

수신되는 진동수는 다음과 같다.

$$\nu = \nu_0\sqrt{\frac{1 - \dfrac{v}{c}}{1 + \dfrac{v}{c}}} = \nu_0\sqrt{\frac{1 - \dfrac{\dfrac{a't^*}{c}}{\sqrt{1 + \left(\dfrac{a't^*}{c}\right)^2}}}{1 + \dfrac{\dfrac{a't^*}{c}}{\sqrt{1 + \left(\dfrac{a't^*}{c}\right)^2}}}} = \nu_0\sqrt{\frac{\sqrt{1 + \left(\dfrac{a't^*}{c}\right)^2} - \dfrac{a't^*}{c}}{\sqrt{1 + \left(\dfrac{a't^*}{c}\right)^2} + \dfrac{a't^*}{c}}}$$

$$= \nu_0\left[\sqrt{1 + \left(\frac{a't^*}{c}\right)^2} - \frac{a't^*}{c}\right]$$

그런데

$$ct^* = \frac{c^2}{a'}\sinh\tau \Rightarrow \frac{a't^*}{c} = \sinh\tau \Rightarrow \nu = \nu_0\,(\cosh\tau - \sinh\tau) = \nu_0\,\mathrm{e}^{-\tau} = \nu_0\left(1 - \frac{ct}{x_0}\right)$$

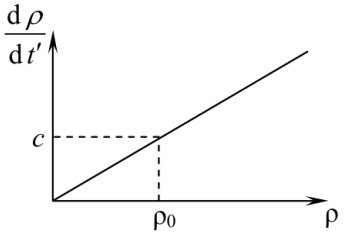
$$t = NT_0 \Rightarrow \nu = \nu_0\left\{1 - \frac{cT_0}{x_0}\left(\left\lfloor\frac{x_0}{cT_0}\right\rfloor + 1\right)\right\}$$

l. **0.5 pt**

$$\frac{\mathrm{d}\rho}{\mathrm{d}t'} = \frac{\mathrm{d}\rho}{\mathrm{d}\tau}\,\frac{\mathrm{d}\tau}{\mathrm{d}t'} = (x_0 - ct)\,\mathrm{e}^{\tau}\,\frac{a'}{c} = \frac{a'}{c}\rho$$

수신 시점에서는

$$\mathrm{e}^{\tau} = \frac{x_0}{x_0 - ct} \Rightarrow \frac{d\rho}{dt'} = (x_0 - ct)\,\frac{x_0}{x_0 - ct}\,\frac{a'}{c} = \frac{c^2}{a'}\,\frac{a'}{c} = c$$



m. 1 pt

$$\tanh\tau = \frac{ct}{x_0} \Rightarrow \frac{1}{\cosh^2\tau}\,\mathrm{d}\tau = \frac{c}{x_0}\,\mathrm{d}t \Rightarrow \mathrm{d}t = \frac{x_0}{c}\left(1-\tanh^2\tau\right)\mathrm{d}\tau = \frac{x_0}{c}\left(1-\frac{c^2t^2}{x_0^2}\right)\mathrm{d}\tau$$

$$\frac{\mathrm{d}\left(\mathrm{d}t\right)}{\mathrm{d}x_0} = \frac{\mathrm{d}\tau}{c}\,\frac{\mathrm{d}\left(\dfrac{x_0^2-c^2t^2}{x_0}\right)}{\mathrm{d}x_0} = \frac{\mathrm{d}\tau}{c}\,\frac{x_0^2+c^2t^2}{x_0^2}$$

$$\varepsilon = \frac{\dfrac{\mathrm{d}\tau}{c}\,\dfrac{x_0^2+c^2t^2}{x_0^2}\,\Delta x_0}{\dfrac{\mathrm{d}\tau}{c}\,\dfrac{x_0^2-c^2t^2}{x_0}} = \frac{x_0^2+c^2t^2}{x_0^2-c^2t^2}\,\frac{\Delta x_0}{x_0}$$

n. 0.5 pt

$$\varepsilon = \frac{\Delta x_0}{x_0} = \frac{h}{\dfrac{c^2}{a'}} = \frac{gh}{c^2} \approx \frac{10\,\mathrm{m/s^2}\cdot 360\cdot 10^3\,\mathrm{km}}{9\cdot 10^{16}\,\mathrm{m^2/s^2}} = 4\cdot 10^{-11}$$

$$\Delta t = 4\cdot 10^{-11}\cdot 365\cdot 24\cdot 3600 \approx 1.26\cdot 10^{-3}\,\mathrm{s}$$

역자 주: 원문에는 $360\cdot 10^3\,\mathrm{km}$로 적혀 있으나, 결과 $4\cdot 10^{-11}$은 $h = 360\cdot 10^3\,\mathrm{m}$ (= 360 km)일 때의 값이다.